Chapter 15 · Semester 2

ELECTRIC CURRENT

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9 worked examples 2 simulations Exercises with answers

15.1 Conduction of electricity

Electric current, I

the **rate of flow of electric charge** through a cross-section of a conductor.

$$I = \frac{\mathrm{d}Q}{\mathrm{d}t} \qquad \text{steady current: } Q = It$$

unit: ampere (A) = C s^{-1}. 1 A is a flow of 1 C of charge per second.

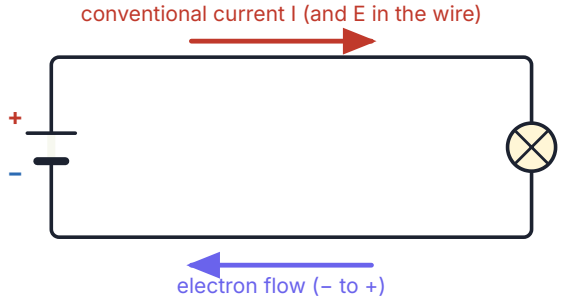


Figure 15.1: the conventional current flows from the positive to the negative terminal outside the cell; electrons flow the other way.

Charge carriers

Material	Charge carriers
Metals	free (conduction) electrons
Semiconductors	electrons and holes (a hole behaves like a positive charge)
Electrolytes and gases (for comparison)	positive and negative ions (and electrons in gases)

How a metal conducts

A metal is a lattice of positive ions with many **free electrons** that are not bound to any atom.

Without a p.d., the free electrons move randomly at high speed (about 10^6 m s^{-1}), colliding with the vibrating ions. Their average velocity is zero, so there is no current.

When a p.d. is applied, an electric field E is set up in the metal. Each electron feels a force eE opposite to the field and accelerates between collisions.

At each collision with an ion the electron loses the velocity it gained. The result is a slow average **drift** opposite to the field, added to the random motion. This drift is the current.

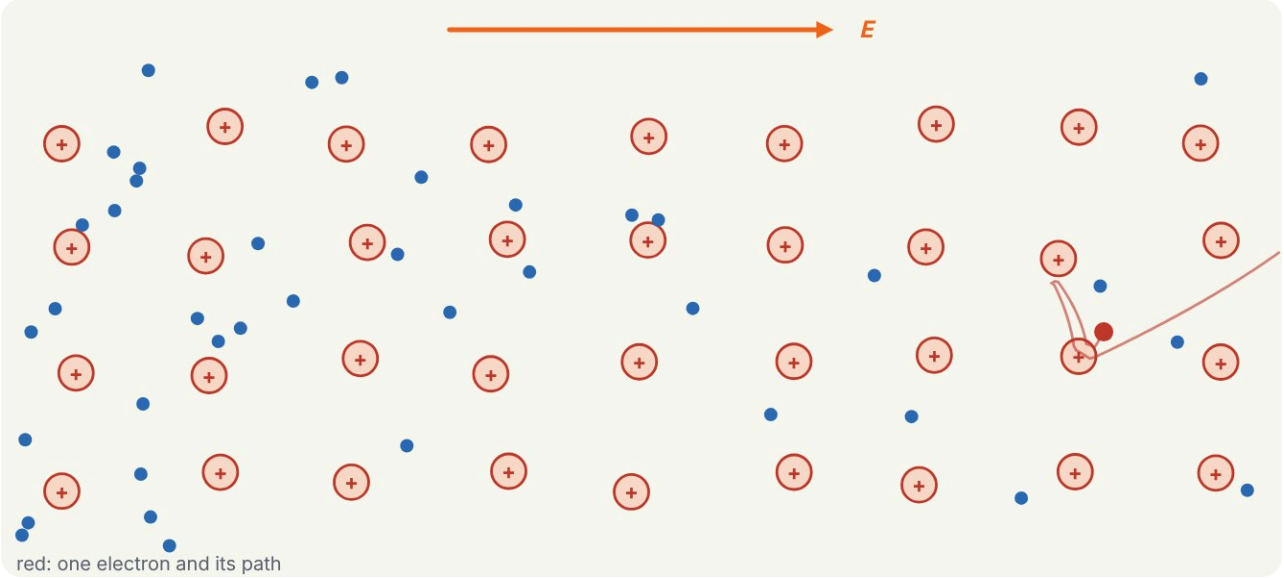


Figure 15.2: the path of a free electron without a field and with a field.

Simulation

Free electrons in a metal

Turn the field on and watch the electrons drift slowly against the field while still moving randomly. Raise the temperature: the ions vibrate more, collisions are more frequent, and the drift becomes slower.



red: one electron and its path

drift (average velocity)
41 units to the left

collisions per second
49

mean time between collisions t
819 ms (simulated)

what to notice
drift is tiny compared with the random speed

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.1.1

A silver wire carries a current of 3.0 A. Find (a) the number of electrons passing a cross-section each second, (b) the charge that flows through a cross-section in 55 s. (e = 1.60 × 10^{-19} C)

Solution

(a) $N = \dfrac{I}{e} = \dfrac{3.0}{1.60 \times 10^{-19}} =$ **1.9×10^{19} electrons per second**

(b) $Q = It = (3.0)(55) =$ **165 C**

15.2 Drift velocity

Drift velocity, v

the **average velocity of the charge carriers along the conductor** when an electric field is applied.

- For electrons it is opposite to the field (and to the conventional current).
- It is very small (about 10^{-4} m s^{-1}) compared with the random speed of the electrons. A lamp lights at once because the field is set up along the whole wire almost immediately and all the electrons start drifting together.

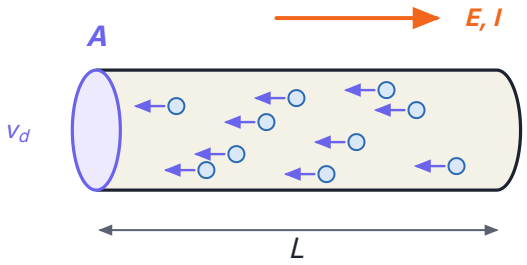


Figure 15.3: a conductor of cross-sectional area A and length L.

Deriving I = Anev

Let n = number of free electrons per unit volume. The volume of the length L is AL, so it holds $N = nAL$ free electrons.

Total charge of these electrons: $Q = Ne = nALe$

They all pass through the end face in the time $t = \frac{L}{v}$

$$I = \frac{Q}{t} = \frac{nALe}{L/v} \Rightarrow I = Anev$$

$$I = Anev$$

n = number density of free electrons (m^{-3}); copper: $n \approx 8.5 \times 10^{28}$ m^{-3}

- For the same current, a thinner wire (smaller A) has a larger drift velocity.
- A semiconductor has a much smaller n, so for the same current its carriers drift much faster.

Example 15.2.1

A copper wire of cross-sectional area 1.0 mm^2 carries a current of 2.0 A. Copper has 8.5×10^{28} free electrons per m^3. Find the drift velocity.

Solution

$$v = \frac{I}{Ane} = \frac{2.0}{(1.0 \times 10^{-6})(8.5 \times 10^{28})(1.60 \times 10^{-19})} = \mathbf{1.5 \times 10^{-4}\ m\,s^{-1}}$$

Example 15.2.2

A transmission line of diameter 3.00 cm and length 100 km carries a steady current of 1500 A. The copper has 8.49×10^{28} free electrons per m^3. How long does one electron take to travel the full length of the line?

Solution

$$A = \pi(0.0150)^2 = 7.07 \times 10^{-4}\ m^2$$

$$v = \frac{1500}{(7.07 \times 10^{-4})(8.49 \times 10^{28})(1.60 \times 10^{-19})} = 1.56 \times 10^{-4}\ m\,s^{-1}$$

$$t = \frac{L}{v} = \frac{1.00 \times 10^{5}}{1.56 \times 10^{-4}} = \mathbf{6.4 \times 10^{8}\ s}\ \text{(about 20 years)}$$

15.3 Current density

Current density, J

the **current per unit cross-sectional area** of a conductor, the area being perpendicular to the current.

$$J = \frac{I}{A} = nev$$

vector, in the direction of the conventional current; unit A m^{-2}

Example 15.3.1

A copper wire carries 10.0 A. Its cross-section is a square of side 2.0 mm. Copper has 8.49×10^{28} free electrons per m^3. Find (a) the current density, (b) the drift velocity.

Solution

(a) $J = \dfrac{I}{A} = \dfrac{10.0}{(2.0 \times 10^{-3})^2} = \mathbf{2.5 \times 10^{6}\ A\ m^{-2}}$

(b) $v = \dfrac{J}{ne} = \dfrac{2.5 \times 10^{6}}{(8.49 \times 10^{28})(1.60 \times 10^{-19})} = \mathbf{1.8 \times 10^{-4}\ m\,s^{-1}}$

15.4 Electrical conductivity and resistivity

Resistance and Ohm's law

Resistance, R

the ratio of the potential difference across a conductor to the current in it: $R = V/I$. Unit: ohm (Ω) = V A^{-1}.

Ohm's law

The current in a metallic conductor is directly proportional to the potential difference across it, provided the temperature and other physical conditions are constant.

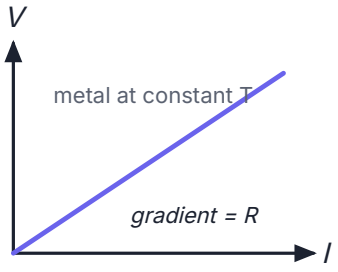


Figure 15.4: an ohmic conductor gives a straight line through the origin.

Resistivity and conductivity

Resistivity, ρ

of a material: the resistance of a sample of unit length and unit cross-sectional area. It depends on the material and its temperature, not on its shape.

$$\rho = \frac{RA}{l} \qquad \left(R = \frac{\rho l}{A}\right)$$

unit: Ω m. Good conductors have a small ρ (copper 1.7×10^{-8} Ω m); insulators a very large ρ.

Electrical conductivity, σ

the **ratio of the current density to the electric field strength** in a conductor.

$$J = \sigma E \qquad \sigma = \frac{1}{\rho}$$

unit of σ: $\Omega^{-1}\,m^{-1}$ ($S\,m^{-1}$). Conductivity is the reciprocal of resistivity.

Ohm's law and J = σE are the same law

For a uniform conductor of length l and area A: $V = El$ and $I = JA$.

Ohm's law $V = IR$ with $R = \rho l/A$: $El = (JA)\frac{\rho l}{A} = J\rho l$

So $E = \rho J$, i.e. $J = \frac{1}{\rho}E = \sigma E$. A constant σ (at constant temperature) means $J \propto E$, which is Ohm's law for the material.

Deriving σ = ne²t/m

Between collisions, a free electron in the field E has acceleration $a = \frac{eE}{m}$.

Let t be the **mean time between collisions**. Each collision wipes out the velocity gained, so the average (drift) velocity gained is $v = at = \frac{eEt}{m}$.

$$J = nev = ne\frac{eEt}{m} = \frac{ne^2t}{m}E$$

Compare with $J = \sigma E$: $\sigma = \frac{ne^2t}{m}$ and $\rho = \frac{m}{ne^2t}$

$$\sigma = \frac{ne^2t}{m} \qquad \rho = \frac{1}{\sigma} = \frac{m}{ne^2t}$$

n = free electrons per unit volume; t = mean time between collisions; m, e = mass and charge of an electron

Temperature and the resistivity of a metal

In a metal, n does **not** change with temperature: every atom already gives its free electrons.

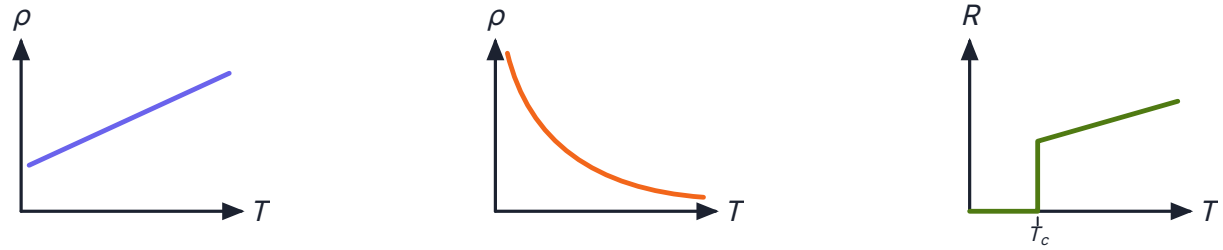
When the temperature rises, the ions vibrate with larger amplitude, so the free electrons collide with them more often: the mean time between collisions t **decreases**.

From $\sigma = ne^2t/m$, σ decreases, so the resistivity $\rho = m/ne^2t$ **increases** (roughly linearly) with temperature.

Superconductivity

the **property of some materials whose resistivity falls to zero when they are cooled below a critical temperature, T_c.**

Material	When the temperature rises	Reason
Metal	ρ increases	n constant; more collisions with the vibrating ions, so t decreases
Semiconductor	ρ decreases (quickly)	more electrons and holes are released, so n increases greatly; this outweighs the fall in t
Superconductor	ρ = 0 below T_c; above T_c it behaves like a normal conductor	mechanism not required

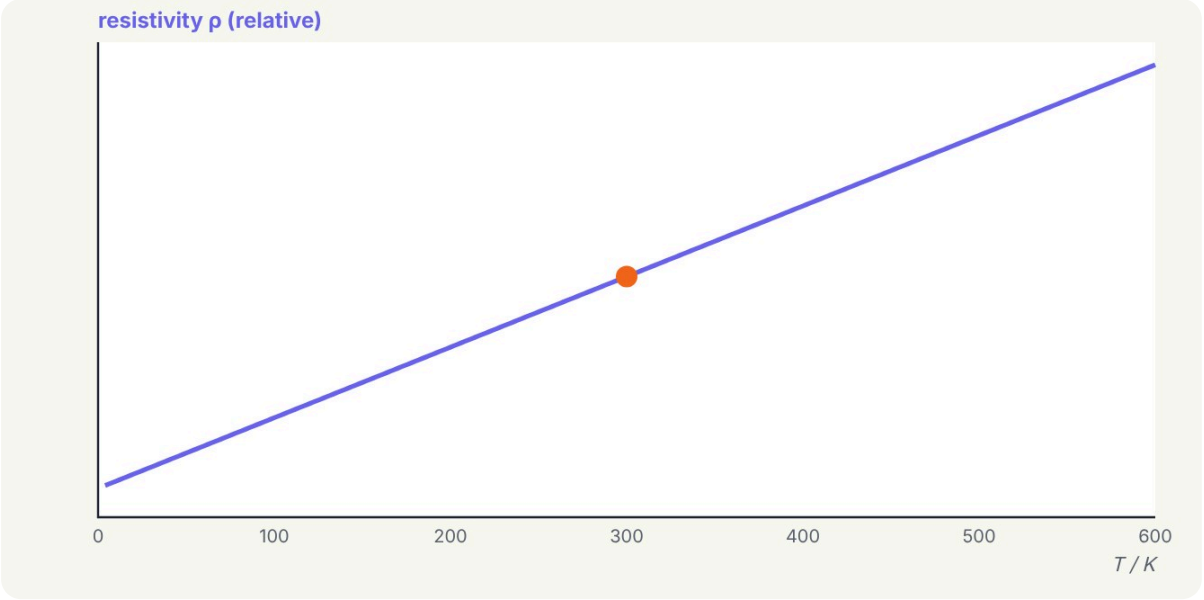


metal semiconductor superconductor

Figure 15.5: how resistivity (or resistance) varies with temperature.

Simulation Resistivity and temperature

Choose a material and change the temperature. Watch the number of carriers n and the mean time between collisions t, and see how they decide the resistivity.



carriers per unit volume n
constant

mean time between collisions t
1.00 × the value at 300 K

resistivity ρ = m/ne²t
1.00 × the value at 300 K

note
curves are sketches, not data

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.4.1

A wire 2.0 m long with a diameter of 0.50 mm has a resistance of 0.171 Ω. Find (a) the resistivity, (b) the conductivity of the material.

Solution

$A = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}\ \text{m}^2$

(a) $\rho = \dfrac{RA}{l} = \dfrac{(0.171)(1.96 \times 10^{-7})}{2.0} = \mathbf{1.68 \times 10^{-8}\ \Omega\,m}$

(b) $\sigma = 1/\rho = \mathbf{5.95 \times 10^{7}\ \Omega^{-1}\,m^{-1}}$

Example 15.4.2

A p.d. of 240 V is applied across a wire 200 cm long. The current density in the wire is 7.14×10^9 A m^{-2}. Find (a) the resistivity, (b) the conductivity of the wire.

Solution

$E = V/l = 240/2.00 = 120$ V m^{-1}

(a) $\rho = \dfrac{E}{J} = \dfrac{120}{7.14 \times 10^9} = \mathbf{1.68 \times 10^{-8}\ \Omega\ m}$

(b) $\sigma = 1/\rho = \mathbf{5.95 \times 10^7\ \Omega^{-1}\ m^{-1}}$

Example 15.4.3

Copper has 8.5×10^{28} free electrons per m^3 and a resistivity of 1.7×10^{-8} Ω m. Find the mean time between collisions of the free electrons. ($m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C)

Solution

$$t = \frac{m}{ne^2\rho} = \frac{9.11 \times 10^{-31}}{(8.5 \times 10^{28})(1.60 \times 10^{-19})^2(1.7 \times 10^{-8})} = \mathbf{2.5 \times 10^{-14}\ s}$$

Example 15.4.4

The copper wire in Example 15.3.1 (10.0 A, square cross-section of side 2.0 mm) is 50 m long. ρ = 1.68×10^{-8} Ω m. Find (a) the electric field in the wire, (b) the p.d. across it, (c) its resistance.

Solution

(a) $E = \rho J = (1.68 \times 10^{-8})(2.5 \times 10^6) = \mathbf{0.042\ V\ m^{-1}}$

(b) $V = El = (0.042)(50) = \mathbf{2.1\ V}$

(c) $R = V/I = 2.1/10.0 = \mathbf{0.21\ \Omega}$ (check: $\rho l/A = (1.68 \times 10^{-8})(50)/(4.0 \times 10^{-6}) = 0.21\ \Omega$)

Section A practice

The resistivity of a wire depends on its

A. length

B. material

C. cross-sectional area

D. shape

Answer

Answer: B

Resistivity is a property of the material (and its temperature). Changing the length or area changes the resistance, not the resistivity.

Exercise: electric current

1. Two wires A and B of the same material and length carry the same current. The diameter of A is twice that of B. Find the ratio of the drift velocities $v_A : v_B$.

   ### Answer

   $v \propto 1/A \propto 1/d^2$, so $v_A : v_B = 1 : 4$

   FULL SOLUTION

   I = nAev, so for the same current and the same material (same n): v ∝ 1/A

   $A = \pi d^2/4 \propto d^2$, and $d_A = 2d_B \Rightarrow A_A = 4A_B$

   $v_A/v_B = A_B/A_A = 1/4$

   $v_A : v_B$ = **1 : 4**

2. Explain why the resistance of a metal wire increases when it gets hotter, but the resistance of a semiconductor decreases.

   ### Answer

   Metal: n stays the same but the ions vibrate more, so the mean time between collisions t falls; $\sigma = ne^2t/m$ falls and ρ rises. Semiconductor: heating frees many more charge carriers, so n rises greatly; this outweighs the small fall in t, so ρ falls.

   FULL SOLUTION

   Conductivity $\sigma = ne^2t/m$, where n is the number of charge carriers per unit volume and t the mean time between collisions; $\rho = 1/\sigma$.

   Metal: n is already very large and does not change with temperature. When heated, the lattice ions vibrate with larger amplitude, so electrons collide more often and t decreases. σ falls, so **ρ (and R) increases**.

Semiconductor: heating gives more electrons enough energy to break free, so n increases greatly. This increase is much larger than the small decrease in t, so σ rises and **ρ (and R) decreases**.

3. An aluminium wire of resistivity 2.8×10^{-8} Ω m and cross-sectional area 2.0 mm^2 carries 4.0 A. Find the electric field in the wire.

Answer

$J = 4.0/2.0 \times 10^{-6} = 2.0 \times 10^{6}$ A m^{-2}; $E = \rho J = 0.056$ V m^{-1}

FULL SOLUTION

A = 2.0 mm^2 = 2.0×10^{-6} m^2

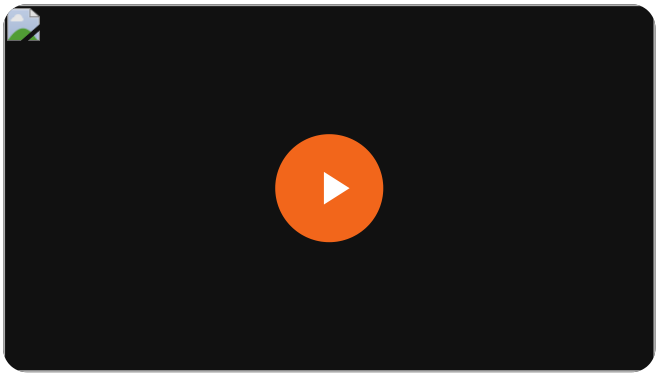
Current density J = I/A = 4.0 / (2.0×10^{-6}) = 2.0×10^{6} A m^{-2}

J = σE = E/ρ ⇒ E = ρJ = $(2.8 \times 10^{-8})(2.0 \times 10^{6})$

= **0.056 V m^{-1}**

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



DEMONSTRATION

How to choose the right fuse: explained with demonstrations · fizzics.org

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Choosing the right fuse

Current heats a wire; a thin fuse wire with a large current density gets hot fastest and melts, breaking the circuit before the cables overheat.